BB84 Quantum Key Distribution — Tutorial Agenda

QCE26 Tutorial · BB84 Quantum Key Distribution: From Photon Polarization to Qiskit Simulation

2 Sessions · 1.5 hr each 30 mins break between sessions Total guided time: 3 hr

> **Flexible attendance:** Each session is self-contained. Attendees may join **Session 1 only**, **Session 2 only**, or **both**. A brief recap at the start of Session 2 re-anchors attendees who missed Session 1.

Track key: THEORY HARDWARE DEMO QISKIT BRIDGE

Session 1 — Protocol, Optical Experiment & Intro to Simulation

Duration: 90 min · Prerequisite: none · Bring: laptop optional

TIME	DUR	TRACK	TOPIC & CONTENT
13:00 – 13:10	10 min	THEORY	**1.1 Classical Cryptography Overview** · Symmetric encryption and key distribution challenges · Asymmetric cryptography and public key infrastructure · Current security assumptions and computational complexity
13:10 – 13:20	10 min	THEORY	**1.2 Quantum Computing Threat Landscape** · Shor's algorithm and RSA vulnerability · Grover's algorithm impact on symmetric cryptography · Timeline for quantum threat realization
13:20 – 13:35	15 min	THEORY	**1.3 Quantum Mechanics Primer for QKD** · Polarization of light: horizontal, vertical, diagonal, anti-diagonal

TIME	DUR	TRACK	TOPIC & CONTENT
			· Measurement bases: rectilinear (+) and diagonal (×) · Quantum measurement and state collapse · No-cloning theorem and its security implications · Heisenberg uncertainty principle in measurement
13:35 – 13:45	10 min	THEORY	**1.4 BB84 Protocol Walkthrough** · Bit encoding, basis selection, transmission, measurement · Basis reconciliation over classical channel · Key sifting process · Error detection and privacy amplification · Security proof intuition: eavesdropper detection via QBER
— 5 min transition —			
13:50 – 14:00	10 min	HARDWARE	**2.1 Experimental Setup Introduction** · Components: linear polarizers, PBS, HWP, photodiodes · Raspberry Pi integration and custom UI · Light source and detection system · Electrical circuit: photodiode voltage generation, ADC conversion

TIME	DUR	TRACK	TOPIC & CONTENT
14:00 – 14:25	25 min	DEMO	**2.2 BB84 Protocol Implementation — Clean Run (No Eve)** · **Alice:** random bit + basis generation, polarization encoding, UI display · **Bob:** random basis selection, HWP basis change, PBS + photodiode measurement, ADC conversion, UI results · **Classical channel:** basis reconciliation, key sifting, shared key generation · **XOR demo:** encrypt sample message with generated key; decrypt to verify · Dashboard: QBER < 11% — key accepted
14:25 – 14:30	5 min	BRIDGE	**Session 1 Wrap-up & Preview** · Recap: protocol steps and clean optical path confirmed · Preview Session 2: Eve attack, QBER spike, Qiskit simulation

30 mins break

Session 2 — Eavesdropping Attack, QBER Analysis & Qiskit Simulation

Duration: 90 min · Prerequisite: none (recap provided) · Bring: laptop with Qiskit + IBM Quantum account (optional but recommended)

TIME	DUR	TRACK	TOPIC & CONTENT
15:00 – 15:10	10 min	THEORY	**Recap + Eavesdropping Theory** · 5-min recap of BB84 for first-time attendees · No-cloning theorem: why copying qubits is physically impossible · The two-hop optical path (Alice → Eve → Bob) · Expected QBER from eavesdropping: ~25% from basis mismatch probability

TIME	DUR	TRACK	TOPIC & CONTENT
15:10 – 15:40	30 min	QISKIT	**3.1–3.2 Qiskit BB84 Circuit Construction** · Verify environment: installations, IBM Quantum account access · Qubit initialization: $\|0\rangle$ and $\|1\rangle$ as BB84 bits · Basis encoding: Z-basis (rectilinear) vs. X-basis (diagonal) · Hadamard gates for basis rotation · Measurement in computational and Hadamard bases · Visualize circuits; analyse outcome probabilities · IBM noise model: decoherence naturally generates a measurable QBER
15:40 – 16:10	30 min	DEMO	**2.3 Eavesdropping Attack — Physical Demonstration** · Run optical experiment — confirm QBER < 11%, key accepted · Introduce Eve and Run optical experiment — QBER > 11%, key rejected · What changes in the bit-error rate and why · QBER formula: tracing the ~25% error rate back to basis mismatch probability
16:10 – 16:30	20 min	THEORY	**Extensions, Outlook & Q&A** · **E91 protocol** · **Real-world QKD deployments and distance limits** · **Open Q&A — hardware, code, Qiskit circuits**

SESSION	FOCUS	HARDWARE NEEDED	LAPTOP NEEDED	STAND-ALONE?
Session 1	Theory + Clean optical run	RPi bench, clean path	Optional	Yes
Session 2	Eve attack + Qiskit + Extensions	RPi bench + ESP32 stage	Recommended	Yes